Splunk SOC Detection Queries

10 practical starting-point SPL searches for Windows, Active Directory, endpoint and web-server investigations

Before you run these searches

Assumptions: Windows Security data is in `index=wineventlog`, the event ID field is `EventCode`, and Sysmon data is in `index=sysmon`. Replace these with your actual index, sourcetype and field names.

Important: These are starting examples, not a validated production detection pack. Field extractions, audit policy, thresholds and time windows must be tested in your environment. Run each search in Search first; review false positives and tune exclusions before creating alerts.

Detection overview

#	Detection	Primary telemetry
1	Failed logins then success	4624, 4625
2	Privileged AD group change	4728, 4732, 4756
3	Suspicious process execution	4688
4	Password spraying	4625
5	Kerberoasting indicators	4769
6	DCSync-like activity	4662 + directory auditing
7	Lateral movement	4624, 4688
8	Services / scheduled tasks	4697, 4698, 4699, 4702
9	Audit log / policy tampering	1102, 4719
10	Web server spawning a shell	Sysmon EventCode 1

1. Failed logins followed by a successful login

```
index=wineventlog EventCode IN (4624,4625)
| eval user=coalesce(TargetUserName, Account_Name, user)
| eval outcome=if(EventCode=4625,"failure","success")
| where isnotnull(user) AND user!="-"
| sort 0 _time
| streamstats time_window=2m
    sum(eval(outcome="failure")) as failures
    by user
| where outcome="success" AND failures>=4
| table _time user failures Computer
```

Purpose: Find a successful logon for an account with at least four failures in the preceding two minutes. Validate event ordering and account-field mappings before production use.

2. Privileged Active Directory group changes

```
index=wineventlog EventCode IN (4728,4732,4756)
| eval group=lower(coalesce(TargetUserName, Group_Name, group))
| eval actor=coalesce(SubjectUserName, Account_Name, user)
| eval member=coalesce(MemberName, Member_Name)
| where match(group,"domain admins|enterprise admins|schema admins|administrators")
| table _time Computer actor group member EventCode
| sort - _time
```

Purpose: Find additions to sensitive groups. Confirm extracted fields, review the actor and added member, and verify whether the change was approved.

3. Suspicious process execution

```
index=wineventlog EventCode=4688
```

```
| eval proc=lower(coalesce(NewProcessName, New_Process_Name, process_name))
| eval parent=lower(coalesce(ParentProcessName, Creator_Process_Name))
| eval cmd=lower(coalesce(ProcessCommandLine, CommandLine, process))
| where match(proc,"(powershell|pwsh|cmd|mshta|rundll32|regsvr32|wscript|cscript)\.exe$")
    OR match(cmd,"-enc(odedcommand)?\\s|frombase64string|downloadstring")
| table _time Computer SubjectUserName proc parent cmd
| sort - _time
```

Purpose: Surface potentially suspicious script and LOLBin execution. Tune using parent process, command line, user, host and approved admin activity. Command-line logging must be enabled.

4. Password spraying

```
index=wineventlog EventCode=4625
| eval src=coalesce(IpAddress, Source_Network_Address, src_ip)
| eval user=coalesce(TargetUserName, Account_Name, user)
| where isnotnull(src) AND src!="-" AND isnotnull(user) AND user!="-"
| bin _time span=5m
| stats count AS failures dc(user) AS accounts_targeted
    values(user) AS accounts values(Computer) AS hosts
    by _time src
| where accounts_targeted>=10 AND failures>=15
| sort - failures
```

Starting threshold: at least 10 distinct accounts and 15 failures per source in five minutes. Tune to your environment and consider distributed attempts across multiple IPs.

5. Kerberoasting indicators

```
index=wineventlog EventCode=4769
| eval user=coalesce(TargetUserName, Account_Name, user)
| eval service=coalesce(ServiceName, service_name)
| eval enc=lower(coalesce(TicketEncryptionType, Ticket_Encryption_Type))
| where enc="0x17" OR enc="23" OR enc="0x0017"
| stats count AS ticket_requests dc(service) AS unique_services
    values(service) AS services values(Computer) AS hosts
    by user
| where ticket_requests>=10 OR unique_services>=5
| sort - ticket_requests
```

Purpose: Hunt for unusual service-ticket request volume and RC4 encryption (commonly 0x17 / decimal 23). Validate field format and baseline; a match is not proof of an attack.

6. DCSync-like directory replication activity

```
index=wineventlog EventCode=4662
| eval props=lower(coalesce(Properties,""))
| where match(props,"replicating directory changes all|replicating directory changes in filtered set|1131f6ad|1131f6ac")
| eval actor=coalesce(SubjectUserName, Account_Name, user)
| stats count values(ObjectName) as objects
    values(Properties) as permissions
    values(SubjectUserSid) as actor_sid
    by _time Computer actor
| sort - _time
```

Broad hunting query only. For production, validate replication permission GUIDs and access masks, exclude legitimate domain controllers/synchronisation accounts, and enable appropriate Directory Service Access auditing/SACLs.

7. Lateral movement — network logon plus process activity

```
index=wineventlog (EventCode=4624 OR EventCode=4688)
| eval user=coalesce(TargetUserName, SubjectUserName, Account_Name, user)
| eval host=coalesce(Computer, ComputerName, dest)
| eval logon_type=coalesce(Logon_Type, LogonType)
| eval process=coalesce(NewProcessName, New_Process_Name)
| stats min(_time) as first_seen max(_time) as last_seen
    values(EventCode) as events values(IpAddress) as source_ips
    values(process) as processes values(logon_type) as logon_types
    by user host
| where mvfind(events,"4624")>=0 AND mvfind(events,"4688")>=0
| table first_seen last_seen user host source_ips logon_types processes events
| sort - last_seen
```

Correlation lead, not proof of sequence: this checks whether both event types exist for the same user and host in the search range. Stronger logic should correlate logon ID and event timing.

8. Persistence — services and scheduled tasks

```
index=wineventlog EventCode IN (4697,4698,4699,4702)
| eval details=lower(coalesce(Message, TaskContent, ServiceName, CommandLine, ""))
| where EventCode=4697 OR EventCode=4699 OR EventCode=4702
    OR match(details,"powershell|cmd\.exe|wscript|cscript|mshta|rundll32|http://|https://|\\\\temp\\\\|\\\\public\\\\")
| table _time EventCode Computer SubjectUserName
    ServiceName TaskName TaskContent details
| sort - _time
```

Events include service installation (4697), task creation (4698), deletion (4699) and task update (4702). Correlate creation and deletion by host/task name and time for a stronger sequence analytic.

9. Security control tampering

```
index=wineventlog EventCode IN (1102,4719)
| eval actor=coalesce(SubjectUserName, Account_Name, user)
| table _time EventCode Computer actor SubjectDomainName Message
| sort - _time
```

Event 1102 records the Security audit log being cleared; 4719 records a system audit policy change. Validate actor, host and whether the action was authorised.

10. Web server spawning a shell

```
index=sysmon EventCode=1
| eval parent=lower(coalesce(ParentImage, ParentProcessName, parent_process_name))
| eval child=lower(coalesce(Image, NewProcessName, process_name))
| eval cmd=coalesce(CommandLine, ProcessCommandLine, process)
| where match(parent,"(w3wp|httpd|apache|nginx|tomcat|php-cgi)(\.exe)?$")
    AND match(child,"(cmd|powershell|pwsh|bash|sh|wscript|cscript|mshta)\.exe?$")
| table _time Computer User parent child cmd
| sort - _time
```

Requires Sysmon process-creation data and appropriate field extractions. A web server spawning a shell is suspicious but may have approved explanations; investigate related inbound requests, file changes and outbound connections.

Production readiness checklist

☐ Confirm index, sourcetype and EventCode/EventID mappings.

☐ Verify required fields are extracted and populated.

☐ Run against historical data and review known-good activity.

☐ Set severity, alert owner and an investigation playbook.

☐ Tune thresholds, approved admin activity and duplicate suppression.

☐ Test with an authorised simulation before enabling production alerting.

Recommended tuning notes

Use correlation searches for sequence-based detections. The failed-login query should validate that failures occurred before the success; lateral movement should correlate the logon session (for example, logon ID) and timing. For DCSync, use relevant replication permission GUIDs/access masks and exclude legitimate replication identities. For web-server compromise, correlate process telemetry with web access logs, file changes and outbound connections.

Use the Splunk Security Content detection library and MITRE ATT&CK; mappings as references when validating and maintaining detections.